The Concept Map

What the system is, where the edge lives, and the maths still missing

Second edition — rewritten around the Scottish lower-league programme

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The goal, stated properly

THE OBJECTIVE

Not to extract large sums from Scottish League Two. **To demonstrate that a systematic pipeline** — model → posterior → staking vector → rebalance — **produces positive out-of-sample log-growth in a market soft enough to actually win in**, and which can then be ported to markets with size.

That ordering is the strategy. Prove the concept where the opposition is weak; scale it where the money is. Low liquidity is not a bug here — **it is the reason the mispricing survives**. There are no sharks in Scottish League Two, which is precisely why the price is wrong.

Each concept states **the maths**, **why your system needs it**, **where you stand** (built / partial / gap / closed), and **an exercise**. Do the exercises by hand. The point is not to produce code — you can already produce code. The point is to be able to tell when the code is wrong.

Part I — The three objects everything is built from

1 The score matrix

What it is. A discrete distribution $P[i, j]$ over the final score. *Everything* downstream is a functional of it.

The maths.

$$P[i, j] = \mathbb{P}\big(X_T = i,\ Y_T = j \mid \mathcal{F}_t\big), \qquad \sum_{i,j} P[i, j] = 1.$$

It must be the **posterior predictive** — averaged over MCMC draws — not the matrix evaluated at the posterior mean. The growth objective is concave, so plugging in the mean **overstakes**.

Why *you* need it. It is the interface between modelling and betting. The engine *produces* it; every market price is a linear functional *of* it; the staking vector is an optimisation *over* it.

BUILT

`compute_score_matrix` → `compute_market_probs`; O/U priced through `SmileScoreMatrix`. Coherence imposed by the IPF tilt in `staking_real`.

2 Markets are linear functionals — so your bets are correlated by construction

The maths. Contract k pays on a set of cells A_k:

$$\pi_k = \sum_{(i,j)\in A_k} P[i, j], \qquad \mathrm{Cov}\big(\mathbf{1}_{A_k},\, \mathbf{1}_{A_l}\big) = \pi_{k\cap l} - \pi_k \pi_l.$$

Why *you* need it. Over 1.5 / 2.5 / 3.5, BTTS and 1X2 are *nested, overlapping sets on one distribution.* They are not independent bets. This is why summing per-bet Kelly gave a **217% joint stake** and a -143% match. **The correlation never needed estimating — it was already sitting in P.**

BUILT

Diagnosed, and solved: the unified solver stakes the whole book jointly on the 144-state grid.

3 The payoff-by-outcome vector — the state variable you are missing

The single most important new idea in this document. When you rebalance over time, *stakes are not the state variable* — because **you cannot unwind a bet at the price you took it**. Laying off Over at minute 70 does not cancel your pre-game Over back; it *locks in* a position, because the odds moved. Stakes placed at different times are not fungible.

The maths. What *is* additive across time is the **net payoff under each outcome**:

$$\boxed{\pi(\omega) \;=\; \sum_{\text{past trades } t} \; \sum_{k} a_{t,k}\, r_k\big(\omega;\; o_{t,k}\big)}$$

a vector in $\mathbb{R}^{169}$ (one entry per score cell), where $r_k(\omega; o)$ is the net return of contract k under outcome ω **at the odds you actually got**.

Why *you* need it. This is what makes in-play rebalancing tractable, convex and correct. Track $\pi(\omega)$, not a stake vector, and the rebalancing program (Concept 7) falls out immediately.

GAP

Not built. `staking_real` optimises a stake vector at one point in time. The moment you rebalance, you need $\pi(\omega)$.

EXERCISE — derive it, then check it against your data

Back Over 2.5 at 1.90 pre-game. A goal goes in; the price moves to 1.45. Now lay Over 2.5 at 1.50 for the same stake.
Write out $\pi(\omega)$ across all outcomes. Confirm it is *not* zero — you have locked a profit, not cancelled a position. **That non-zero vector is exactly why stakes are the wrong state variable.**

Part II — Where the edge is, and why

4 The three-way asymmetry (the strategic core)

	The market	**Your model**
Supremacy $(\lambda_h - \lambda_a)$	*Efficient* — headline market, deep money	*Weak* — needs team strength / xG
Totals level (λ_{tot})	*Efficient* — easy to price	*Fine* — but no edge over them
Totals *shape* $(\varphi(K))$	**Crude** — one λ through a Poisson	**Strong** — the smile pillar

CLOSED — do not revisit

There is no 1X2 edge. 17 cells in `split_market_pillar`; home ROI negative in every one; all $p > 0.25$. Your own verdict: *"Don't chase 1X2."* Settled.

Note that CURATED05 ($w = 0$ on 1X2) winning the staking race is not a workaround — it is the *correct* response to a market you cannot beat.

The corollary is the good news. What xG and player ratings buy you is a better *supremacy* estimate — they improve the one axis where you have no edge anyway. **So losing them in Scotland costs you nothing that matters.**

5 The smile: your edge, and what generates it

What it is. The market prices the O/U ladder by pushing a *single* intensity through a Poisson CDF. Reality is not Poisson. The per-strike deviation is your edge.

The maths.

$$\mathbb{P}\big(N_{\text{tot}} \le K\big) \;=\; \text{cdf}\Big(\text{Poisson}\big(\lambda_{\text{tot}} \cdot \varphi(K)\big),\; K\Big), \qquad \varphi \equiv 1 \iff \text{Poisson.}$$

$\varphi(K)$ is the **implied-volatility smile of football**. Exactly as Black–Scholes prices every strike with one σ and the market's implied σ then varies by strike, a Poisson book prices every line with one λ and the implied λ varies by strike.

Why *you* need it. All of your measured edge lives here. `li_smile50`: BTTS GLMEdge $p = 0.01$, top pooled-totals coefficient. And it is **market-inverted** — it consumes the de-vigged O/U ladder, **not xG**. Your own note: *"the smile machinery is rate-source-agnostic."* **This is why the Scottish move works, and why a richer Bet365 ladder (0.5–7.5 on ∼99% of matches) is an *upgrade* on what you had.**

The structural hypothesis — and a one-parameter smile

Connect two of your own findings. You measured residual overdispersion var/mean ≈ 1.09 *and* a home/away remaining-goal correlation of $+0.207$. **One mechanism generates both**: a shared match-level frailty θ ("is this an open, end-to-end game?") multiplying both intensities.

The maths. With $N_{\text{tot}} \mid \theta \sim \text{Poisson}(\theta\,\bar\lambda)$ and $\theta \sim \Gamma(a, a)$, the total is Negative Binomial and (law of total variance)

$$\frac{\text{var}}{\text{mean}} \;=\; 1 + \frac{\bar\lambda}{a} \qquad \Longrightarrow \qquad a \;=\; \frac{\bar\lambda}{\text{var}/\text{mean} - 1} \;\approx\; 28.9 \qquad (\bar\lambda = 2.6,\; \text{v/m} = 1.09).$$

Now invert that Negative Binomial through a Poisson, strike by strike — and a smile appears for free:

strike K	0.5	1.5	2.5	3.5	4.5	5.5
$\varphi(K)$ predicted by the frailty	0.958	0.974	0.990	1.006	1.022	**1.038**

It is a monotone **skew**, not a U-shape — Poisson is bounded below at zero, so the left tail has nowhere to go and a Gamma mixture fattens the *right* tail only. Read it as: the market's Poisson *understates* $\mathbb{P}(N > 4.5)$ and *overstates* $\mathbb{P}(N > 0.5)$, so **high Overs and low Unders are both underpriced by a Poisson book.**

THE TEST THAT MATTERS — one plot, and it is cheap

Plot the $\varphi(K)$ your Scottish grid *estimates* against the $\varphi(K)$ the frailty *predicts*.

- **They match** ⇒ your smile *is* overdispersion. You can then replace a free-form $\log\varphi \in \mathbb{R}^{K_{\max}+1}$

(5–6 free parameters) with **one parameter** a. In a thin league where every free parameter costs you, that is a large win — and it **generalises**, because a is a property of football, not of one league's odds.

- **Your φ exceeds the prediction** $\Rightarrow$ the excess is genuine market mispricing *beyond* the overdispersion, and **the gap $\varphi^{\text{est}} - \varphi^{\text{frailty}}$ is your edge, isolated.**
- **Your φ is U-shaped, not monotone** $\Rightarrow$ the frailty is *not* the mechanism, and something more interesting is happening. Also worth knowing.

Either way you have decomposed φ into "what a known structural effect explains" and "what is left" — and the residual is what you are actually being paid for.

EXERCISE — derive it, then check it against your data

Derive $\mathrm{var}/\mathrm{mean} = 1 + \bar{\lambda}/a$ for the Gamma–Poisson mixture (law of total variance; four lines).
Reproduce the $\varphi(K)$ table: for each K, solve $\text{Poisson-sf}(K; \Lambda) = \text{NegBin-sf}(K)$ for Λ, and set $\varphi = \Lambda/\bar{\lambda}$.
Then overlay your grid's estimated φ. That single plot is the most informative thing you can produce this month.

Part III — Staking: the convex core

6 Log-optimal growth over the outcome space

The maths.

$$f^{\star} \;=\; \arg\max_{f} \sum_{i,j} P[i,j]\, \log\Big(1 + \sum_{k} f_k\, r_k(i,j)\Big).$$

Concave in f (log of an affine map) $\Rightarrow$ unique global optimum; a small convex program.

BUILT

`staking_real`: coherent IPF grid tilt $\rightarrow$ unified solver on the 144-state grid, capped at $\Sigma \le 0.2$.

READ YOUR OWN RACE TABLE

strategy	term W	G/match $\pm$ SE	maxDD	ruined
`U_cap02` (raw unified, w=1)	**0.01**	-0.0169	1.00	**YES**
`TRUST_EB_U_cap02` (learned w)	2.98	$+0.0040$	0.87	–
`CURATED05_U_cap02` (w=0 on 1X2)	**26.86**	$\mathbf{+0.0120 \pm 0.0040}$	0.42	–

The optimiser is not your problem. Feed it good probabilities and it compounds $\times 26.9$ at $t \approx 3$. Feed it the 1X2 probabilities and it takes you to zero — because a joint Kelly with *negative edge* on a line will stake it into the ground, and **no cap saves you from that**. Sizing was never the issue. **Per-line edge quality** is.

7 Rebalancing with transaction costs — the in-play staking layer

This is the piece to build. Using $\pi(\omega)$ from Concept 3, at each state change, with the current model matrix P_t and the current odds o_t:

The maths.

$$\boxed{\Delta a^\star \;=\; \arg\max_{\Delta a} \underbrace{\sum_\omega P_t(\omega)\,\log\Big(W_0 + \pi(\omega) + \sum_k \Delta a_k\, r_k\big(\omega;\, o_{t,k}\big)\Big)}_{\text{growth of the \textit{total} position}} \;-\; \underbrace{c \sum_k |\Delta a_k|}_{\text{crossing cost}}}$$

Why *you* need it. Three things fall out **for free**, and each replaces a hand-tuned rule you currently run:

1. **It stays convex.** A concave log term minus a convex ℓ_1 term. Unique optimum, milliseconds.
2. **The no-trade region appears automatically.** The subgradient of $|\Delta a_k|$ at 0 is $[-c,\, c]$, so the optimiser *refuses to trade* unless the growth gradient exceeds the crossing cost. **You never need a rule for *when* to rebalance — the cost term *is* the rule.** That is Davis–Norman, obtained simply by writing the objective down correctly.
3. **Hedging happens by itself, for the right reason.** If a goal makes your Over position likely to win, the optimiser lays Over — not because you told it to, but because log is concave, so *equalising wealth across outcomes raises* $\mathbb{E}[\log W]$. Your grid-searched $\varphi \approx 0.5$–0.75 hedge fraction is this optimiser's answer. It will also correctly *decline* to hedge when the price moved against you — which a fixed-φ rule cannot.

And note: **this is equally correct pre-game**, where Δa at kickoff is the only trade. So build it against the Scottish pre-game book *now*. In-play rebalancing is then the same function, called again.

GAP

Not built. `basic_hedging` has a rule-based φ at a fixed minute; `staking_real` has a single-shot solve. Neither tracks $\pi(\omega)$, and neither prices the crossing cost.

Read. Davis & Norman (1990), *Portfolio selection with transaction costs*, *Mathematics of Operations Research* 15(4). Read the *idea* of the no-trade region, not the viscosity solutions.

8 Risk-constrained Kelly — replacing your cap

The maths.

$$\max_f \; \mathbb{E}\big[\log(1 + r^\top f)\big] \qquad \text{subject to} \qquad \mathbb{P}\big(\text{drawdown beyond } \alpha\big) \;\le\; \beta.$$

Busseti–Ryu–Boyd derive a **convex bound** on the drawdown probability, so the constrained problem stays solvable, controlled by a single interpretable risk-aversion parameter.

Why *you* need it. Your winner runs a **maxDD of 0.42**. You are currently stacking *two* ad-hoc risk controls — fractional Kelly *and* a $\Sigma \le 0.2$ cap. This replaces both with one derived constraint, and their headline result is that it **beats fractional Kelly at the same drawdown risk**.

GAP

Not built. The cheapest real improvement available to you.

Read. Busseti, Ryu & Boyd (2016), *Risk-Constrained Kelly Gambling*, arXiv:1603.06183. **If you read one paper, read this one.**

9 Parameter uncertainty — and why it matters MOST in a thin league

The maths.

$$f^\star = \arg\max_f \mathbb{E}_{\theta\sim\text{post}}\Big[\mathbb{E}_{\omega|\theta}\big[\log(1+f^\top r)\big]\Big] \neq \arg\max_f \mathbb{E}_{\omega|\bar\theta}\big[\log(1+f^\top r)\big].$$

Average over the posterior *inside* the log.

Why *you* need it. This is your strongest argument for a full Bayesian engine — stronger than you have been pitching it. In Scottish L2 (goals only, no xG, five seasons) your posterior on λ is **wide**. A point-estimate Kelly would *massively* overstake, because the growth objective is concave and Jensen punishes you for ignoring parameter uncertainty. **The posterior is worth most precisely where the data is thinnest.** It is not a nice-to-have in a lower league — it is the thing that stops your own parameter noise from ruining you, and it is where nearly everyone else blows up.

BUILT

`BayesianKelly` (Baker–McHale); the L2 calibrator shifts the whole MCMC posterior, not a scalar.

CARRY IT THROUGH

When you move to the rebalancing problem, the optimisation must average over posterior **draws** of the score matrix, not use the mean matrix. It would be very easy to silently lose the uncertainty here.

Part IV — In-play

10 Transient versus structural mispricing

	Transient	**Structural**
Why the price is wrong	Mid-adjustment; stale quotes; post-goal turmoil	The market's *model* is wrong
How long it stays wrong	Seconds to minutes	**The rest of the match**
What it takes to capture	**Speed** — low latency, colocation	**A better model**
Competing against	Bots	A crude repricing rule
Is it yours?	**No**	**Yes**

Your $+29\%$ ROI on stale fills *was* the transient mispricing, and you correctly called it fake (*"the model knows a goal the stale quote hasn't yet absorbed"*). With realistic fills it collapsed to $+0.8\%$. **The mispricing was real and you could not have it.**

Your thesis is correct, and you have the evidence

The market must reprice a *high-dimensional* object — λ as a function of (score, minutes left, man advantage) — under time pressure, roughly 500 times a season. The pre-game price gets *days* of arbitrage. So the market's *settled* in-play price should be systematically worse. Two independent pieces of evidence say it is:

	Team that just scored	Team that conceded
Market-implied, $\mathbb{Q}$ (Vecer et al.)	0.61	**1.01** — *no response*
Your realised fit, $\mathbb{P}$	$e^{-0.24} = 0.79$	$\mathbf{e^{+0.25} = 1.28}$

The market prices **no response at all** for the trailing team. Your data — paired CV, $t = 3.15$ — says they score **28% more**. That is not turmoil; it is a *crude repricing rule*, and it stays wrong for the remaining hour. **Nobody has to be beaten to the punch.**

And independently: your in-play model **beats the market's own inverted** λ_{rem} out-of-sample (-1.0245 vs -1.0352). That is a *measurement* of structural in-play mispricing, not a hypothesis.

11 The one number that settles in-play

YOU CANNOT CURRENTLY COMPUTE IT

There is **no matched-volume, size, bid or ask column anywhere in your historical Betfair data — for any league.** LTP is a *print*, not a *quote*. So your $+0.8\%$ is not a number; it is a guess whose error bars you cannot compute. **Every in-play backtest you own is unfalsifiable for execution.**

The maths. The whole in-play question reduces to one comparison, per bin:

$$\boxed{\underbrace{\text{edge}_t \;=\; p_t^{\text{model}} \times o_t^{\text{available back}} \;-\; 1}_{\text{at the REAL price, not LTP}} \qquad \text{versus} \qquad \underbrace{\text{size available at } o_t}_{\text{can you fill a real stake?}}}$$

Is $\mathbb{E}[\text{edge}_t] > 0$ **at sizes above your minimum stake**?

Why *you* need it. Your structural edge might be 2% and the in-play spread 3% — in which case the edge is real, correctly measured, and **unprofitable**. That gap *is* the difference between your $+29\%$ and your $+0.8\%$, and neither number is trustworthy.

The experiment is already running. `betfair_live.order_book_1m` carries bid/ask prices *and* volumes and is collecting on Ireland. Let it accumulate. In four weeks you can compute the box above, and in-play is settled permanently in one direction or the other. **Cost: zero.**

CLOSED — do not revisit

Scottish in-play is dead. From the liquidity audit: half of Ireland's tick rate; p90 gap 4–4.7 min (Ireland: 2 min); p99 up to 14 minutes of dead air; **a full 1X2 in only 44% / 38% of in-play bins** (Ireland: 83%) — below your own 50% threshold, so the inversion is *unidentified* most of the time; and 20% of L2 matches carry no market at all. **Do not trade Scottish in-play, and do not buy data to try.** Scotland is a *pre-game* market.

Part V — The closed file

SETTLED. DO NOT RE-OPEN. (I have proposed each of these to you at least once.)

1. **No 1X2 / supremacy edge.** 17 cells; home ROI negative in all; $p > 0.25$ throughout.
2. **No latent in-play state to learn.** A match has ≈ 2.7 goals. With the largest latent effect your data supports (sd 26%), watching an *entire half* of football reduces the posterior sd by **1%**; halving it would need ≈ 33 matches of exposure. **Therefore: no filtering, no SDEs, no particle filters, no Kushner–Stratonovich.** Your observable-covariate regression is the correct design, and this is the reason why.

   Corollary — on Zou et al.: their scheme "works" only by faking a weak prior. Setting $r_1 = E_H(45)$ means a goalless first half **halves** a team's scoring rate — exactly 0.500, by construction, for every team, in every match. The honest update is -5%. **They over-react by a factor of ten**, and their own RPS table shows the advantage *reversing* by half-time. Read their returns as an Under-bias harvested from a soft Asian book, not as evidence that in-play Bayesian updating works.

3. **No transient in-play edge.** Execution latency is binding; the exchange reprices faster than you can act.

4. **Books you do not need:** Shreve; Brigo–Mercurio; Wu; Lando; Särkkä; West & Harrison; Brémaud; the SMC/`Turing.jl` paper. All were on my earlier lists. All are ruled out by (2).

Reading list

Tier 0 — this week ($\approx$ 80 pages total)

1. **Busseti, Ryu & Boyd (2016)**, *Risk-Constrained Kelly Gambling*, arXiv:1603.06183. **Read first** — it is your staking problem, exactly.

2. **Cover & Thomas**, *Elements of Information Theory*, **Ch. 6 only**. Multi-outcome Kelly — and the theorem that your growth-rate edge *equals the mutual information* between your model and the outcome. **Your edge, measured in bits** — a far better headline metric than backtest ROI, which your own work showed is contaminated.

3. **Vecer, Ichiba & Laudanovic (2006)**, §2.1–2.2 only. The goal and red-card multipliers.

4. **Davis & Norman (1990)**. Skim for the no-trade region.

Tier 1 — the core, over a couple of months

5. **Boyd & Vandenberghe**, *Convex Optimization* (free PDF). Ch. 2–4 plus the log-optimal investment examples. You need to *recognise* convexity, not prove theorems.

6. **MacLean, Thorp & Ziemba**, *The Kelly Capital Growth Investment Criterion*. Editors' introduction + Thorp's survey.

7. **McElreath**, *Statistical Rethinking* (2nd ed). **The antidote to LLM-built code** — it teaches you to *build* a model rather than recognise one. You said you were worried you are not learning; this is the book that fixes that.

8. **Baker & McHale (2013)**, *Optimal betting under parameter uncertainty*. You already use it — now read *why* it works.

Order of work.
(1) Finish the Scottish grid — it is live, and it is the only thing earning.
(2) Plot your estimated $\varphi(K)$ against the frailty prediction (Concept 5). One plot, decisive.
(3) Build the payoff-vector rebalancer (Concepts 3 + 7) — correct pre-game and *in-play.*
(4) Risk-constrained Kelly (Concept 8), to kill the 0.42 *drawdown.*
(5) Let the Ireland order book accumulate; then compute edge-versus-spread (Concept 11).
Everything else is closed.